



DEIVA MANI KAILASAM

Senior Software Engineer (Full Stack, GenAI)

Cloud-Native Distributed Systems | GenAI & Agentic AI Applications

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🌐 [LinkedIn: deivamani-k] 📁 [Portfolio]

SKILLS

• Languages & Application Stack:

Python, TypeScript,
Node.js/Express, Java, Spring
Boot, JavaScript, React, Angular,
Next.js, C#

• GenAI & Agentic AI:

LLMs, RAG
pipelines, embeddings, prompt
engineering, Vector DBs, MCP
tools & integrations, LangChain,
LangGraph, LlamaIndex,
Snowflake Cortex, LLM-powered
analytics, multi-agent
orchestration, AI-assisted
developer workflows

• Frontend & UI Engineering:

Tailwind CSS, SCSS, PrimeNG,
Angular Material, AG Grid
Enterprise, Highcharts,
Storybook, Compodoc, MDX,
white-label theming, design
systems

• APIs, Data & Messaging:

FastAPI,
REST, gRPC, GraphQL,
OpenAPI/Swagger, PostgreSQL,
SQL Server, Snowflake, MongoDB,
MongoDB Atlas Search, Kafka,
Redis, Caffeine, JSONB, full-text
search, Liquibase

• Cloud, DevOps & Platform:

AWS
(ECS, Fargate, EC2, Lambda, EKS,
Bedrock), Azure (App Service,
Functions, PostgreSQL, SQL,
Cosmos DB, Blob Storage, AD
B2C, Azure DevOps), Docker,
Kubernetes, Terraform, Helm,
ArgoCD, ARM/Bicep, Jenkins,
GitHub Actions, Git

PROFILE

Senior Software Engineer (Full Stack, GenAI) with 8+ years of experience building GenAI-enabled, cloud-native, multi-tenant SaaS platforms on AWS and Azure using Python, TypeScript, Node.js, Java/Spring Boot, React, Angular, and Next.js. Delivered production systems combining intelligent backend services, interactive data-rich frontends, and strong platform engineering foundations, including distributed APIs handling 10K+ daily requests and products serving 100K+ users. Specialized in GenAI and agentic applications using RAG, tenant-scoped embeddings, LLM-powered analytics, LangChain, LangGraph, LlamaIndex, and Snowflake Cortex, deployed to production via Docker and Kubernetes (AWS EKS). Brings end-to-end ownership across architecture, implementation, testing, CI/CD, observability, and mentoring with a focus on reliability, performance, and business impact.

WORK EXPERIENCE

Kinesso, Omnicom Media Group

AUG 2021 - PRESENT

Senior Software Engineer (Full Stack)

GenAI & Agentic Platforms

- Designed and implemented scalable Python FastAPI services and data pipelines for multi-tenant GenAI applications, integrating RAG pipelines with tenant-specific embeddings stored in MongoDB Atlas vector search and handling 10K+ daily inference requests with sub-200ms latency.
- Designed GenAI pipelines for tenant-aware embeddings, LLM-based anomaly detection, and a Python-based data quality scoring engine, reducing undetected data quality issues around 75%.
- Developed agentic AI prototypes using LangChain and LangGraph for multi-agent orchestration and document-querying workflows integrated with production data systems, partnering with marketing and analytics stakeholders to validate outputs against business KPIs.
- Built a Snowflake analytics agent using LangGraph and Snowflake Cortex for natural-language querying and dynamic dashboard generation, enabling business users to explore data and surface analytics insights through interactive dashboards.
- Established LLMops workflows for production deployment of fine-tuned GPT-4 variants using Docker containerization and Kubernetes on AWS EKS.

Backend, APIs & Data Platforms

- Architected and led a multi-module Spring Boot 3.x microservices platform with JWT/OAuth2, multi-tenant data access, and OpenAPI contracts, serving 10K+ daily API requests across PostgreSQL, SQL Server, Snowflake, and MongoDB.
- Designed multi-tenant isolation patterns using DB-per-tenant, collection-per-tenant, and row-level security, ensuring zero cross-tenant leakage across 12+ environments.
- Implemented a gRPC-based plugin architecture for integrating Python microservices and client-specific React dashboard plugins, improving onboarding speed for new services.
- Built a hybrid data access layer spanning PostgreSQL, SQL Server, Snowflake, and MongoDB with Liquibase migrations and MongoDB schema versioning, simplifying cross-database querying.

- **Engineering Practices:** Test automation, Pytest, JUnit, Jest, Cypress, Playwright, MockMvc, clean code, code review, performance tuning, incident response, Datadog, Application Insights, Copilot, Cursor AI, Claude, Agile
- **Architecture & Design:** System design, microservices, distributed systems, multi-tenant SaaS, API design, CQRS, Saga, reactive programming, SQL/NoSQL data modelling, test-driven development

NOTABLE PROJECTS

- **Snowflake Agent:** Built a Python-based analytics agent using LangGraph and Snowflake Cortex for natural-language querying and dynamic dashboard generation over Snowflake data, supporting analytics workflows with 85% prediction accuracy.
- **AI Agent for Code Generation:** Built an LLM-powered internal engineering tool using GPT-4 and LangChain to generate boilerplate and integration code, accelerating delivery for internal teams.
- **Feature-Rich UI Component Library:** Built a multi-framework design system across Angular, React, and Next.js with 40+ reusable components used across multiple internal products.
- **Edge Water Management Platform:** Delivered a full-stack web application for smart water monitoring and optimization using Angular and cloud backend services.

- Integrated Google Ads, Meta/Facebook, Bing Ads, Adverity, Snowflake, Power BI, and MongoDB Atlas Search using OAuth2/OIDC and Python data connectors, enabling near real-time advertising analytics.
- Reduced 99th percentile latency by about 55% through Spring WebClient optimization, Python asyncio, MongoDB GridFS, parallel query execution, and hybrid caching using Caffeine, Redis, and MongoDB.

Frontend & Product Experience

- Built and maintained a multi-framework design system with 40+ reusable components across React, Angular, and Next.js, distributed through private npm and adopted across 6 internal products.
- Implemented advanced AG Grid Enterprise tables with filtering, sorting, row expansion, and CSV/Excel export through ExcelJS, enabling non-technical users to self-serve analytics.
- Delivered a white-label theming system across React, Next.js, and Angular supporting 5+ brand variants with dynamic stylesheet injection.
- Authored Storybook documentation with 150+ stories, accessibility tests, and AI-assisted component suggestions, improving component discoverability by 85%.

Platform, Quality & Leadership

- Established automated testing across 1000+ unit tests and 100+ integration/E2E tests using Jest, Vitest, Pytest, Cypress, Playwright, and MockMvc, achieving 80%+ coverage across Java, Python, and UI layers.
- Configured CI/CD with Jenkins and GitHub Actions for React/Next.js using multi-stage Docker builds, SonarQube gates, and pull-request validation, cutting merge-to-deploy time by 40%.
- Designed Kubernetes infrastructure for Spring Boot, Python FastAPI, and Next.js workloads using ConfigMaps, Vault secrets, MongoDB StatefulSets, health probes, and Datadog dashboards, reducing MTTR by around 65%.
- Migrated CI/CD pipelines to hybrid AWS environments using Terraform and ArgoCD, automating Docker and Kubernetes deployments of Python AI services with Helm charts, Kyverno policy-as-code, and reusable GenAI library modules.
- Developed Terraform infrastructure for AWS ECS/Fargate and MongoDB Atlas with GitOps via ArgoCD, enabling one-command deployments and eliminating manual infrastructure changes.
- Mentored 5+ engineers, with 3 promoted to mid-level roles, and drove coding standards across Python, TypeScript, Java, and multi-service codebases.

ACE Contractors Pvt Ltd Software Developer

DEC 2020 – JUL 2021

- Architected and owned a Progressive Web App platform for construction operations on Azure App Service, Azure PostgreSQL Flexible Server, Azure Functions, Blob Storage, and AD B2C, achieving 99.9% uptime for 200+ daily active users across 10+ sites.
- Built a mobile-first Angular frontend with PWA capabilities including Service Workers, App Manifest, and Cache API, enabling offline workflows for field workers and improving daily task completion rates by 40%.
- Implemented Node.js/Express backend services with gRPC microservices and stream processing for real-time timesheet validation and contract updates, achieving sub-100ms response times for 95th percentile requests.
- Designed PostgreSQL schemas for employees, projects, contracts, timesheets, and safety incidents using JSONB and full-text search for efficient retrieval.
- Delivered CI/CD using Azure DevOps YAML pipelines, multi-stage Docker builds, Azure Container Registry, and ARM/Bicep templates, reducing deployment time from 2 hours to 12 minutes.
- Led weekly executive syncs with the company owner, prioritizing high-ROI features that reduced payroll disputes, digitized timesheets, and accelerated management documentation and reporting workflows by 3x.

EDUCATION

- **Master of Technology in Information Technology**
La Trobe University
Jul 2019 – May 2021
- **Bachelor of Technology in Information Technology**
Valliammai Engineering College
Aug 2011 – May 2015

CERTIFICATIONS

- Google Cloud Certification on **GenAI** (2025)
- IBM Certification on **Prompt Engineering** (2025)
- **Best IT Project** Award – La Trobe University (2020)
- **Pursuit of Knowledge** Award, **On the Spot** Award, **Cornerstone** Award, **Star** Award – TCS (2016–2018)

Tata Consultancy Services

JAN 2016 – MAY 2019

Software Developer (Full Stack) / Systems Engineer

- Led migration of a legacy Java EE monolith to Spring MVC and Spring Boot layered architecture, reducing technical debt by 70% and removing 15+ critical security vulnerabilities.
- Gradually migrated AngularJS 1.5 to Angular 5 using RxJS, Angular Router, Angular Material, and lazy loading, achieving 3x faster initial page loads and 65% bundle size reduction.
- Designed PostgreSQL schemas replacing Oracle for projects, employees, timesheets, and billing with stored procedures and indexing, cutting query times by 65% and improving page load performance.
- Implemented test-first development with JUnit, Mockito, TestNG, and Selenium WebDriver E2E testing, achieving 85% unit test coverage and catching 25+ regressions early.
- Containerized Spring Boot services using Docker multi-stage builds, enabling 8+ major releases that consolidated multiple legacy systems.